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Abattoir surveillance: The U.S. experience

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Abstract

Abattoir, or slaughter, surveillance has been an important component of bovine tuberculosis control and eradication programs in the U.S., and has adapted to changes in the livestock market from farm to table, and the threat of bovine tuberculosis from a wildlife reservoir. The purpose of this overview was to describe the current goals of U.S. bovine tuberculosis slaughter surveillance, describe the elements of slaughter surveillance in the U.S., describe enhancements to the slaughter surveillance system, and discuss future challenges for the U.S. bovine tuberculosis surveillance program. Government regulations and the scientific literature were examined to provide information for this paper. The control and eradication of bovine tuberculosis in livestock falls to the United States Department of Agriculture and two agencies within the Department: the Food Safety and Inspection Service (FSIS) and the Animal and Plant Health Inspection Service (APHIS). FSIS conducts routine slaughter surveillance for disease or conditions that render carcasses unsuitable for human consumption, while APHIS is involved in antemortem bovine tuberculosis testing, and necropsy and investigation of bovine tuberculosis cases identified through slaughter surveillance or antemortem testing. Results from the previous 5 years of surveillance are presented. Enhancements have been added to the current surveillance system to improve its performance. An incentive program has been used to increase the numbers of tissues submitted for laboratory examination, the state of Michigan is implementing electronic animal identification under a pilot program, and expansions to the current system are being developed to accommodate new livestock industries. The success of these programs and challenges for the future are discussed.

Keywords: *Mycobacterium bovis*; Abattoir surveillance; Slaughter surveillance; Bovine tuberculosis; *M. bovis*; Epidemiology

1. Introduction

Slaughter surveillance has been an important component of bovine tuberculosis control and eradication programs in the U.S., since the United States Department of Agriculture (USDA) began bovine tuberculosis eradication programs in 1917 (Frye, 1995). In the 1950s, when area testing was becoming an inefficient method to detect bovine tuberculosis when the prevalence was low, the focus of bovine tuberculosis control in the U.S. moved to slaughter inspection, where epidemiological investi-

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gations would be initiated following meat inspectors' identification of infected animals at slaughter. Since this time, slaughter surveillance in the U.S. has adapted to changes in the livestock market from farm to table, the development of new industries providing red meat for consumers, and the threat of bovine tuberculosis from a wildlife reservoir. The objectives of this paper are to (1) describe the current goals of U.S. bovine tuberculosis slaughter surveillance, (2) describe the elements of slaughter surveillance in the U.S., (3) describe enhancements to the slaughter surveillance system, and (4) discuss future challenges for U.S. bovine tuberculosis surveillance programs.

2. Goals of bovine tuberculosis surveillance in the U.S.

The presence of bovine tuberculosis in livestock and wildlife reservoirs creates a threat to public, animal, and economic health of the U.S. population and its livestock industry, and its elimination has been a priority since the turn of the 20th Century. The control and eradication of bovine tuberculosis in livestock falls under the purview of the USDA. One of the primary missions of the USDA is to provide the nation with safe and affordable food, through 17 distinct agencies conducting research, education, marketing, and economic support. These agencies work to support the nation's diverse agricultural industries, sustain the health, diversity and productivity of the Nation's forests and grasslands to meet the needs of present and future generations, and promote public health and well-being by providing nutritional education, and a safe and wholesome food supply.

3. The current USDA slaughter surveillance program for bovine tuberculosis

Surveillance for bovine tuberculosis in the U.S. is conducted through two USDA agencies, the Food Safety and Inspection Service (FSIS) and the Animal and Plant Health Inspection Service (APHIS). The FSIS is the public health agency in the USDA responsible for ensuring that the nation's commercial supply of meat, poultry, and egg products is safe, wholesome, and correctly labeled and packaged. The APHIS protects and promotes agricultural health by administering the Animal Welfare Act and carrying out wildlife damage management activities. All livestock at slaughter are inspected for evidence of bovine tuberculosis by FSIS inspectors, while APHIS is responsible for ante-mortem testing of livestock, and slaughter inspection of animals that are bovine tuberculosis-suspect or may have been exposed to bovine tuberculosis.

The livestock industry in the U.S. is diverse, and provides a variety of red meat for human consumption. The FSIS classifies livestock slaughtered into seven categories (cattle, calves, sheep, goats, swine, equids, and other hoofstock (deer, antelope, buffalo, etc.)), of which the majority is made up of cattle and swine. These animals come from many different sources, from small farms raising fewer than 20 head, to large production units producing tens of thousands of animals each year. The types and sizes of livestock rearing operations vary regionally, with larger operations located primarily in the west and southwestern portions of the country. Many of these animals either the food processing system through sale yards, while others go directly from the rearing or finishing operation to slaughter.

In the U.S., red meat slaughter plants can be classified as either large regional slaughter plants, or small/very small custom slaughter plants that often serve individual consumers rather than larger markets. In fiscal year (FY) 2004, 40 large regional slaughter plants located in 19 states slaughtered nearly 94% of all adult cattle in the U.S. (United States Animal Health Association, 2004). Conversely, many of the small slaughter plants process a variety of species (e.g., goat, bison, deer) for specific regional or ethnic customers. The FSIS conducts routine inspection of slaughter facilities using hazard analysis and critical control points (HACCP) guidelines to avoid contamination of carcasses during processing, to reduce the levels of foodborne pathogens entering the human chain.

3.1. Slaughter surveillance under USDA-FSIS

As previously mentioned, the primary goal of the FSIS is to ensure a safe and wholesome food supply. At slaughter plants throughout the U.S., the FSIS conducts the inspection of carcasses for disease or other conditions, which if present, may result in the animal being condemned as unfit for human consumption. Animals arriving at slaughter may be condemned prior to slaughter for evidence of disease, injuries, or other conditions indicative that the animal may not be appropriate for human consumption. After slaughter, any inspected carcasses not meeting the minimum standard of safety are condemned, and are excluded from the human food chain. The numbers of livestock slaughtered and condemned by FSIS inspectors is considerable. When examining annual slaughter-based statistics for cattle, calves, and other bovids and cervids (Table 1), differences can be seen in the numbers of animals slaughtered, the numbers condemned both pre- and post-mortem, and those condemned for evidence of bovine tuberculosis.

Table 1
Cattle and other bovine and cervids, slaughtered and condemned, including tuberculous cattle, FSIS, 1998–2002

Slaughter category				Condemned		
Year	Main	Sub-category	Number slaughtered	Ante-mortem	Post-mortem	For bovine tuberculosis[a]
1998	Cattle	Bulls and stags	615888	233	1375	0
		Steers	16202791	1380	12764	10
		Cows	5886745	21673	108797	3
		Heifers	10567435	969	9933	4
		Calves	1444625	14859	13799	0
	Others[c]		10582	0	24	21
1999	Cattle	Bulls and stags	601652	750	1127	0
		Steers	16647224	1078	17270	12
		Cows	5342619	21438	91555	5
		Heifers	11088609	769	21288	4
		Calves	1267573	11709	12315	1
	Others[c]		15553	7	16	0
2000	Cattle	Bulls and stags	619616	660	1352	0
		Steers	17457463	−48941[b]	72377	7
		Cows	5269576	25144	117730	0
		Heifers	11789720	954	19638	0
		Calves	1103173	11046	11362	0
	Others[c]		19065	8	12	0
2001	Cattle	Bulls and stags	706958	505	1697	0
		Steers	18363668	1941	15398	24
		Cows	6214474	30756	135125	4
		Heifers	12355710	1274	11527	14
		Calves	1333417	13490	11739	0
	Others[c]		20833	2	26	0
2002	Cattle	Bulls and stags	610130	170	1314	1
		Steers	15621561	1884	10104	37
		Cows	5175861	25815	117669	68
		Heifers	9996325	1253	7662	24
		Calves	1033547	10067	9465	0
	Others[c]		23810	39	109	0

[a] Includes animals from routine slaughter and reactor cattle sent to slaughter.
[b] Inconsistency in numbers from FSIS ADRS data.
[c] Deer, antelope, buffalo, etc.

In addition to federally inspected slaughter inspection through the FSIS, there are currently 28 states operating meat inspection programs, inspecting over 2000 meat and poultry slaughter plants (http://www.fsis.usda.gov/regulations_&_policies/state_inspection_programs/index.asp). These slaughter plants are all small or very small, and the state slaughter inspection programs provide more personalized guidance to these establishments in developing their food safety operations. The FSIS provides funding and training support to these state programs, which are required to enforce standards "at least equal to" those imposed under the Federal Meat Inspection Act and the Poultry Products Inspection Act.

To ensure that slaughter surveillance is capable of detecting bovine tuberculosis at low levels in slaughtered cattle, it is imperative to have adequate numbers of suspicious lesions submitted for examination. Since adoption of the Comprehensive Strategic Plan for the Eradication of bovine tuberculosis in 2000, the rate of submissions of suspicious granulomatous tissues has increased annually (Fig. 1). Reports on slaughter submissions are sent quarterly to FSIS, APHIS, and State Veterinarians to keep all agencies aware of submission rates.

The goal in 2000 was to achieve a rate of five submissions per 10,000 head of adult cattle killed at a given slaughter facility (United States Department of Agriculture, Animal and Plant Health Inspection Service, 2005). Of the 40 slaughter plants mentioned above, 21 (53%) slaughtered 55% of all adult cattle in FY 2004, and had a combined submission rate of 14.1 per 10,000, accounting for 84.9% of all granuloma submissions (United States Animal Health Association, 2004). Of the remaining 19, five accounted for 5.8% of adult cattle slaughtered, and a combined submission rate of 4.0 per 10,000 (2.5% of all submissions). Unfortunately, the remaining 14 plants submitted only 5.2% of all adult submissions at a rate of 1.5 submissions per 10,000 killed, while slaughtering 32.3% of all cattle slaughtered annually. Two of these plants made only one submission each in FY 2004, and efforts are under way to resolve the sampling problems in these two facilities (United States Animal Health Association, 2004). In the first half of FY 2005, 30 of the 40 slaughter plants were submitting at acceptable rates (including four of the 14 lowest submitters from 2004), another four had improved their submission rates to more than 50% of the acceptable rate, but six of the lowest submitters were still submitting suspicious lesions at less than 50% of the acceptable rate (Meyer, 2005). Overall, the number of submissions is increasing, and the numbers of positive samples has remained relative stable for the last two fiscal years (Fig. 2).

3.2. Slaughter surveillance under USDA-APHIS

The primary focus of APHIS in bovine tuberculosis surveillance is on ante-mortem testing, which com-

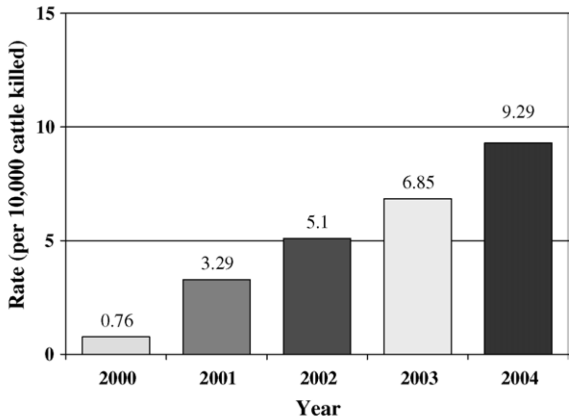

Fig. 1. Rate of submission of granulomatous tissues for laboratory examination from adult cattle through slaughter surveillance, fiscal years 2000–2004.

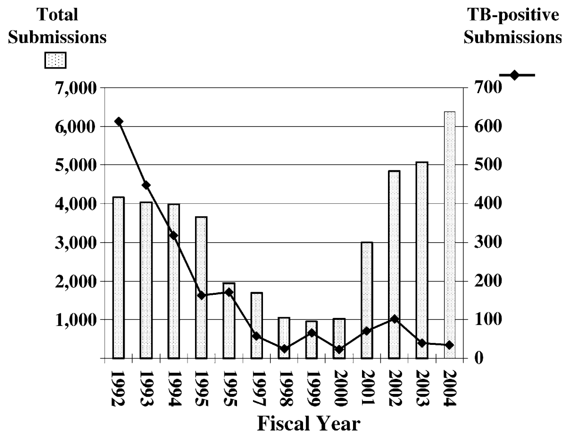

Fig. 2. Numbers of suspicious granuloma submissions and bovine tuberculosis-positive submissions, fiscal years 1992–2004.

plements routine slaughter surveillance conducted by FSIS. Any cattle, goats, cervids, or bison that show a positive response for *Mycobacterium bovis* to the cervical skin test (CT, classified as reactor), or comparative cervical skin test (CCT, classified as reactor) are sent to slaughter within 15 days of classification, where they will be examined by FSIS inspectors, or may be destroyed on the premises or a postmortem examination facility under the supervision of a State or Federal animal health veterinarian to ensure that a proper postmortem examination is conducted (United States Department of Agriculture, Animal and Plant Health Inspection Service, 2005). Animals that respond positively to the caudal fold tuberculin test (CFT, classified as suspect) may be subject to additional testing with a CCT test or a bovine interferon gamma assay to confirm a diagnosis of *M. bovis* infection, or may be sent directly to slaughter. If animals are classified as suspect by the CCT, they may be retested with the CCT in 60 days or be sent directly to slaughter.

All postmortem examinations must be overseen by State or Federal animal health veterinarians (United States Department of Agriculture, Food Safety Inspection Service, 1994). The veterinarian conducts a standard postmortem examination, and includes examination of the supramammary and mesenteric lymph nodes for evidence of bovine tuberculosis. Tissues and samples of any suspicious lesions are sent to the National Veterinary Services Laboratory (NVSL) for analysis. Histopathological examination of the samples is conducted, the samples are classified as negative, suspicious, or compatible with mycobacteriosis, and the results are reported to the veterinarian performing the postmortem examination. At NVSL, polymerase chain reaction (PCR) testing is used on formalin-fixed tissues that are histologically compatible with *M. bovis* infection to detect genetic evidence of infection. The current gold standard for identification of *M. bovis* infection is mycobacterial culture.

The detection of bovine tuberculosis at slaughter initiates an investigation to identify the source of the infected animal. The 'traceback' investigation begins by identifying the immediate source of the infected animal, and the attempts to identify the animal's ultimate farm of origin. Given the infectious nature of bovine tuberculosis, it is critical to identify where the animal may have acquired infection and the location of any other animals that may have been exposed to the tuberculous animal. The success of traceback investigations is highly dependent on the availability of individual animal identification and thorough recordkeeping as the animal progresses from birth to slaughter. For example, in FY 2003, 23% of 39 cases of bovine tuberculosis had no animal identification of any type, making it difficult if not impossible to locate the herd of origin (United States Animal Health Association, 2002). Insufficient record-keeping by owners of a bovine tuberculosis-positive adult beef cow identified in FY 2004 made traceback to herd of origin impossible, despite the fact that the cow did have an official ear tag for identification at slaughter (United States Animal Health Association, 2004). These examples illustrate some of the difficulties encountered when conducting tracebacks, and current estimates of the rate of success of tracebacks are between 50 and 70% of investigations undertaken.

4. Enhancements to the current slaughter surveillance system

Work is ongoing to improve the current bovine tuberculosis slaughter surveillance program in the U.S., with enhancements coming from both national and state agencies. There are examples of programs that have improved the existing surveillance system (the USDA bovine tuberculosis Eradication Performance Awards Program), are working to improve animal identification to improve the rates of successful tracebacks (the National Animal Identification System and the Michigan Electronic Identification Program), or have expanded the surveillance system to encompass a growing industry (the Michigan Farmed Deer/Captive Cervid Slaughter Surveillance Program).

4.1. The USDA bovine tuberculosis eradication performance awards program

The USDA bovine tuberculosis eradication performance awards program was designed to increase the numbers of lesions from adult cattle submitted for laboratory examination, by providing financial incentives for FSIS inspectors and APHIS veterinary medical officers (VMOs) based on results of lesion

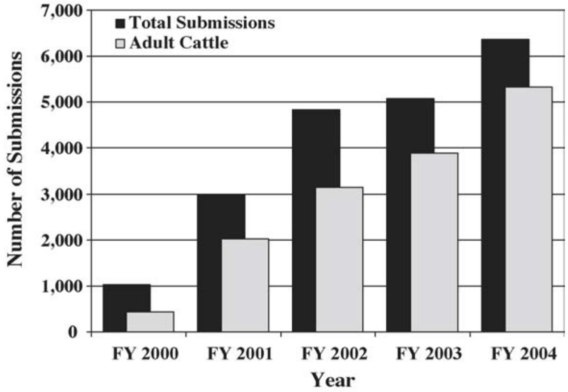

Fig. 3. Number of granuloma submissions, total and from adult animals, FY 2000–2004.

submissions. If a submitted sample is positive by histopathology, an award of $100 U.S. for a steer or $500 U.S. for an adult animal is paid, and the award is doubled if lesion is culture- or PCR-positive for *M. bovis*. If the APHIS veterinary service (VS) epidemiological investigation locates an infected herd, a $6000 U.S. reward is shared by the FSIS inspectors and VMOs on the case. Additionally, high-submitting FSIS slaughter inspection groups can win a performance award of $300 per team member. Recent increases in the numbers of submissions from adult cattle (Fig. 3) have been attributed to this program.

4.2. Animal Identification: the National Animal Identification System and the Michigan Electronic Identification Program

There is a great need for improved animal identification and record-keeping systems, as the rates of successful tracebacks indicate. A brief report by the FSIS, of 42 meat inspectors at slaughter facilities across the U.S., described some of the problems encountered when collecting animal identification at slaughter (United States Department of Agriculture, Food Safety Inspection Service, 2002). Inspectors reported that a wide variety of identification systems were in use, tag and identification requirements differed significantly between states, and plant recordkeeping systems were dependent on the quality of information being received from a variety of sources (e.g., sale barns, feedlots, shippers, producers, auction houses). Larger slaughter facilities had more formalized recordkeeping systems, while very small plants relied on log books or paper invoices for records. The inspectors reported that a significant percentage of animals had no identification (ID), and some plants had difficulty keeping animal ID tags with all parts of the carcass. When animals were handled in lots rather than as individuals, there were difficulties in counting the numbers of animals, and animals were often re-grouped so that it was impossible to identify the source of a given animal from a specific lot. The report made several recommendations, including requiring individual animal identification for all animals entering slaughter, development of a uniform system of identification, clarifying the roles of APHIS, FSIS, and the slaughter plant in tracebacks, the use of electronic tracking and sharing of databases (United States Department of Agriculture, Food Safety Inspection Service, 2002).

The National Animal Identification System (NAIS), administered by USDA/APHIS, is a cooperative state-federal-industry program being developed to track animal movement from birth to death. The major components of the system are premises identification, animal identification, and animal tracking. The information collected and managed through NAIS will make disease surveillance and control programs more efficient and effective, by providing rapid data access and electronic communication of information.

One area that NAIS has focused on is the identification of livestock premises, group/lots, and individual animals. Currently, producers must use different identification numbers for a given animal for a variety of different uses: animal health programs (i.e. brucellosis tag numbers), industry-sponsored programs (herd improvement programs), or interstate commerce. Matching identification numbers between different systems makes tracking the animal from farm to slaughter slow and difficult. The NAIS system will be standardized to use 7-character premises ID numbers, 13-character group/lot numbers, and 15-character animal identification numbers (AIN).

Part of the state of Michigan's ongoing bovine tuberculosis eradication plan is the electronic identification (EID) program, which was launched as a pilot program by the Michigan Department of Agriculture (MDA) and USDA/APHIS in November 2001 (Kirk, 2005). The state of Michigan lost its tuberculosis-free status in 1998 with the identification of a bovine tuberculosis-infected cattle herd in the northeastern lower peninsula of the state, after discovery of endemic bovine tuberculosis in the regions wild white-tailed deer population. Since that time, the state has implemented control and surveillance programs to reduce levels of bovine tuberculosis in the wild deer population and the state's livestock herds (Anon., 2005a). As of 1 January 2005, more than 1,049,000 cattle from over 17,000 herds have been subjected to antemortem testing, and 33 bovine tuberculosis-positive cattle herds have been identified. The state achieved split-state status in 2004, with the north-eastern portion of the lower peninsula classified as modified accredited (MA) by the USDA/APHIS, and the remainder of the state classified as modified accredited advanced (MAA) (United States Department of Agriculture, Animal and Plant Health Inspection Service, 2005).

The Michigan EID program is currently focused on the lower peninsula of the state of Michigan, where the MA area is located. The EID program provides cattle producers in the MA area of the state, or bovine tuberculosis-accredited herds from the rest of the lower peninsula, with radio frequency identification device (RFID) tags free of charge. Each RFID tag is linked to the National Farm Animal Identification and Records Program (FAIR) and the USDA Generic Database system (GDB). Data collected in these systems include animal date of birth, gender, species/type, and bovine tuberculosis test status. As of 31 May 2005, there were 15,644 premises in the database system, of which 1945 were provided with RFID tags. A total of 241,054 RFID tags distributed, and over 120,024 animals have been identified with RFID tags.

The tags are assigned to beef cattle at weaning, and to dairy cattle at calving, and the animal's information is entered into the database. When these animals enter concentration points, such as markets or slaughter plants, the tags are read with a hand-held wand device, and the data are downloaded to a laptop computer for transmission to the main database, along with the date, time, and location of the animal at the time of reading. Using the tag reader eliminates the need for paper recordkeeping, and greatly reduces the time required to update and maintain records. The database compiles the history of each animal entered in the system, as it moves from farm of origin to slaughter, and will improve the tracing of livestock and provide up-to-date information on the numbers and locations of animals in the state. The system is very useful for the issuance of movement permits: one market had a total of 931 permits issued, covering 5339 animals with RFID tags.

Michigan livestock markets and regional slaughter plants receiving Michigan cattle are being equipped with EID RFID tag readers. Eleven of the fourteen livestock markets have been equipped with EID readers, and have read 41,132 animals to date. The seven regional cattle processing plants have read a total of 17,741 cattle at slaughter, and 1837 cattle were read at 25 small custom slaughter plants in Michigan. Plans are under way to equip the remaining markets with EID readers, equip staff with hand-held computers and wireless data transmission facilities so data is easily accessible in the field, and using EID for the state's farmed cervids.

4.3. The Michigan surveillance protocol for slaughter examination of white-tailed deer and elk

The State of Michigan has an expanding captive cervid industry, which has been expanding in the state to over 800 operations raising primarily white-tailed deer and elk. Given the presence of *M. bovis* in the free-ranging white-tailed deer of northeastern Michigan, and the discovery of a bovine tuberculosis-infected captive cervid operation in 1997 (Kaneene et al., 2002), surveillance in captive cervids has been a priority in the state's bovine tuberculosis control program.

The Michigan farmed deer/captive cervid slaughter surveillance program (MDA, 2000) exceeds the current USDA requirements for slaughter surveillance of captive cervids (United States Department of Agriculture, Animal and Plant Health Inspection Service, 2005). The Michigan surveillance program requires examination of all animals 12 months of age or older, and the number of animals to be tested is based on the average adult herd size (Table 2). This table indicates the numbers of animals to be sampled over three consecutive calendar years, with at least 25% of total number examined yearly. Since smaller herds may not be able to meet sampling requirements through slaughter, animals from same herd with negative single cervical skin tests can contribute to

Table 2
Tuberculosis surveillance requirements for captive white-tailed deer and elk hunting ranches

Avg. adult herd size (N)	# To test	Avg. adult herd size (N)	# To test
<46	N	96–100	77
47–49	$N-1$	101–110	81
50–51	49	111–120	84
52–53	50	121–130	88
54	51	131–140	91
55	52	141–150	94
56–60	56	151–160	96
61–65	59	161–170	99
66–70	63	171–180	101
71–75	65	181–190	103
76–80	68	191–200	105
81–85	70	201–225	108
86–90	73	226–250	111
91–95	75	251–275	115
		276 and up	117

total to be sampled, as long as at least 50% undergo slaughter surveillance, and each animal is only tested once during the 3-year cycle. The program has successfully examined over 3700 deer through slaughter surveillance, and no bovine tuberculosis-positive animals have been found (Table 3), and versions of the program are being adopted in other states with captive cervid industries, including Indiana, Minnesota, Montana, New York, and Wisconsin.

All animals examined under surveillance (through slaughter surveillance or antemortem testing) are given an official identification ear tag, and any parts saved for examination must be stored in the same bag or container with ear tag ID, or, if individual parts are stored in bags, each bag must contain a similarly numbered tag to maintain identification through the laboratory testing process. The Michigan program also requires the maintenance of a slaughter surveillance record for privately-owned cervids. These records include the ID number and results of bovine tuberculosis examination for each animal, owner information and location of the farm, and where the examination was conducted. This information must be submitted to the MDA by the examining veterinarian within 10 days of examination.

5. Future challenges for bovine tuberculosis surveillance programs in the U.S.

As described, enhancements are being implemented to improve the current bovine tuberculosis slaughter surveillance program in the U.S., but areas exist where the program can be strengthened.

One important area for bovine tuberculosis slaughter surveillance is the achievement of targets set for sample submission from slaughter plants. The

Table 3
Results from surveillance (cervical skin tests and slaughter) of captive cervids in Michigan, 2000

Species	Test	# Negative	# Suspect/reactor	# Positive
White-tailed deer	Single cervical	3815	318	0
	Comparative cervical	271	13	0
	Slaughter	3393	51	0
Elk	Single cervical	7472	139	0
	Comparative cervical	148	8	0
	Slaughter	250	8	0
Red deer	Single cervical	1261	24	0
	Comparative cervical	16	0	0
	Slaughter	6	0	0
Fallow deer	Single cervical	1036	225	0
	Comparative cervical	144	69	0
	Slaughter	22	0	0
Other cervids[a]	Single cervical	1209	69	0
	Comparative cervical	50	4	0
	Slaughter	16	0	0

[a] Includes reindeer, sika deer, axis deer, mule deer, muntjac, "exotic", and "mixed".

aforementioned incentive program has successfully increased the number of submissions, but there are still regional slaughter plants that have not reached these goals and additional work is needed to bring their submission rates up.

Another area of concern in bovine tuberculosis slaughter surveillance is rates of successful traceback investigations of bovine tuberculosis-positive animals, and the need to improve these success rates from the current 50–70% to a goal rate of 75%. Key to the success of tracebacks is the collection and availability of current and accurate information about individual animals, and the ability to trace these animals from farm of origin to slaughter. This can be achieved through the use centralized or integrated computer databases with standardized data sets that contain information about animal origins, movement, and status. The NAIS system will accomplish these goals, and the U.S. Secretary of Agriculture has issued a timeline for implementation of NAIS (United States Department of Agriculture, 2005). This timeline calls for premises and animal identification to be completed by January 2008, and animal movement tracking to be fully functional by January 2009.

The use of electronic identification systems, such as the system currently used by the Michigan EID program, will also make animal identification easier and less subject errors found in other animal ID systems, and will streamline the data collection process at points of concentration. As the RFID technology improves, the use of hand-held wands may be replaced by readers that can be built into chutes, making the process of collecting large numbers of animal IDs faster and more accurate.

Finally, there is one area in which improvements can be made, and that is through the development of an integrated approach between government agencies to livestock slaughter surveillance for bovine tuberculosis. As described above, the basic mandates of FSIS and APHIS are similar, but differ in fundamental philosophies:

“Because the activities of FSIS inspectors address primarily human food safety risks rather than animal disease risks, APHIS has never been able to rely completely on sample collection by FSIS inspectors to provide all the samples needed for a statistically valid evaluation of the animal disease profile of animals passing through a slaughter plant.”

Federal Register, 2002 (67)229. USDA/APHIS, 9 CFR Part 71, [Docket No. 99–017–1], RIN 0579–AB13, Blood and Tissue Collection at Slaughtering Establishments, Proposed rule.

Currently, slaughter inspection is managed by the FSIS, while much of the effort in tracebacks is borne by APHIS. There are circumstances where communication issues, and the differing agendas between agencies result in different priorities, which can create problems in the field. The Performance Awards Program is an excellent example where both agencies have chosen to prioritize one objective, namely increasing tissue submissions for bovine tuberculosis surveillance, and has demonstrated that this approach can be successful. More inter-agency cooperation can only improve the effectiveness of the existing surveillance program, and can better utilize the farm-to-slaughter data collection and tracking programs planned for the future of U.S. bovine tuberculosis slaughter surveillance.

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